

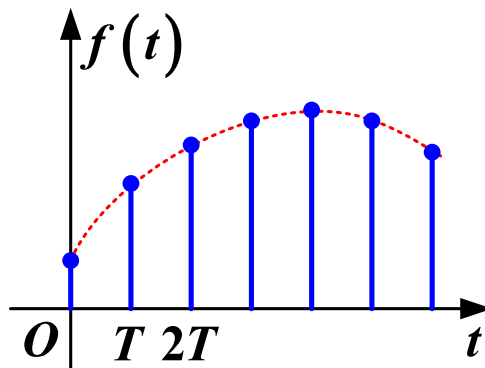
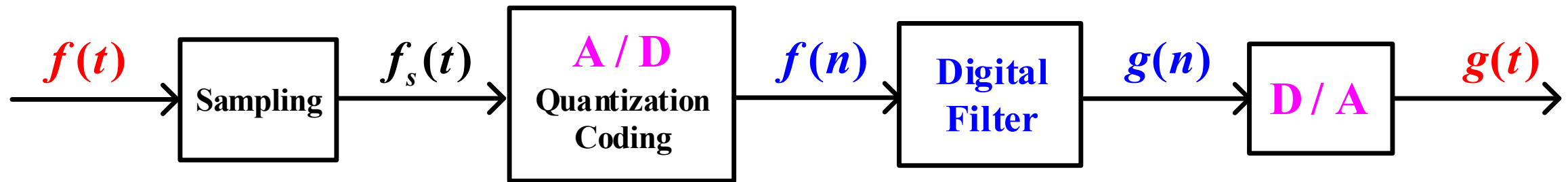
Main Content

- **Concept of Sampling**
- ✓ **Digital Processing of Analog Signals**
- ✓ **Sampling Interval and Sampling Frequency**
- **Spectrum of Sampled Signals**
- **Time-Domain Sampling Theorem**
- **Reconstruction of Continuous-Time Signals from Samples**
- ✓ **Frequency-Domain Filtering**
- ✓ **Time-Domain Interpolation**
- **Aliasing**

4. Sampling Theorem and Aliasing

1. Concept of Sampling

(1) Digital Processing of Analog Signals



$$f_d(n) = f(t)|_{t=nT}$$

T : Sampling interval

$$f_s = \frac{1}{T} : \text{Sampling frequency}$$

Problems to be addressed:

- (1) How to set the **sampling frequency**?
- (2) Can $f(t)$ be **reconstructed** from $f_s(t)$?

(2) How to Set the Sampling Frequency?

Principle of Motion Pictures:

The fundamental principle of film is "*persistence of vision*". Scientific experiments show that after an image disappears, the human eye retains its impression on the retina for approximately **0.1–0.4 seconds**. When film reels advance at **24 frames (samples) per second**, the rapid succession of static images leverages this physiological phenomenon to create a seamless illusion of motion

Key Considerations for Frame Rate Selection:

- Too high a sampling frequency: Introduces data redundancy
- Too low a sampling frequency: Causes motion distortion (e.g., judder or flicker)

4. Sampling Theorem and Aliasing

(2) How to Set the Sampling Frequency?



**Sampling Rate
6kHz**



**Sampling Rate
12kHz**



**Sampling Rate
18kHz**



Impact of Different Sampling Frequencies on Images

4. Sampling Theorem and Aliasing

(2) How to Set the Sampling Frequency?

Mosquito Wingbeat Frequency: ~ 600 Hz



High-Speed Recording, Standard Playback

Sampling Frequency/Frame Rate: Determined by the target frequency characteristics

4. Sampling Theorem and Aliasing

(2) How to Set the Sampling Frequency?

Stellar Rotation Frequency: $1/86,400$ Hz



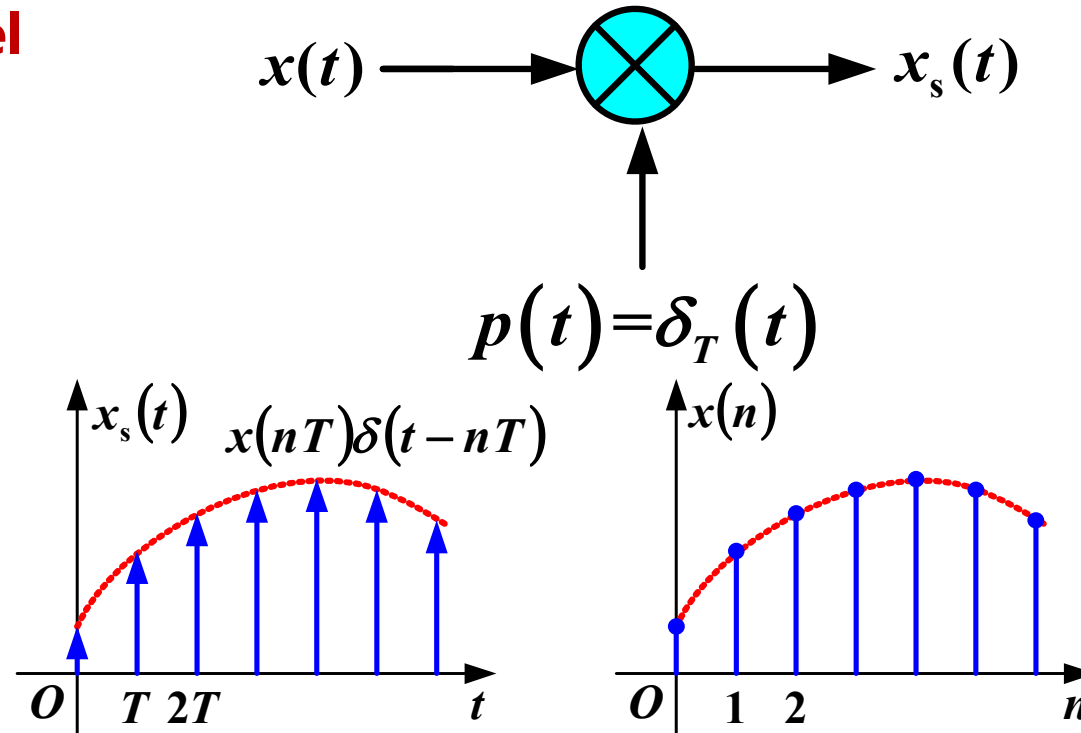
Time-Lapse Recording, Real-Time Playback

Sampling Frequency/Frame Rate: Determined by the target frequency characteristics

4. Sampling Theorem and Aliasing

2. Spectrum of Sampled Signals

(1) Sampling Model



Mathematical Model of Sampled Signals (Continuous-time to Discrete Sequence):

$$x_s(t) = x(t)\delta_T(t)$$

$$x(n) = x(t)|_{t=nT}$$

4. Sampling Theorem and Aliasing

2. Spectrum of Sampled Signals

(2) Spectrum of the Sampled Signal

$$p(t) = \delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

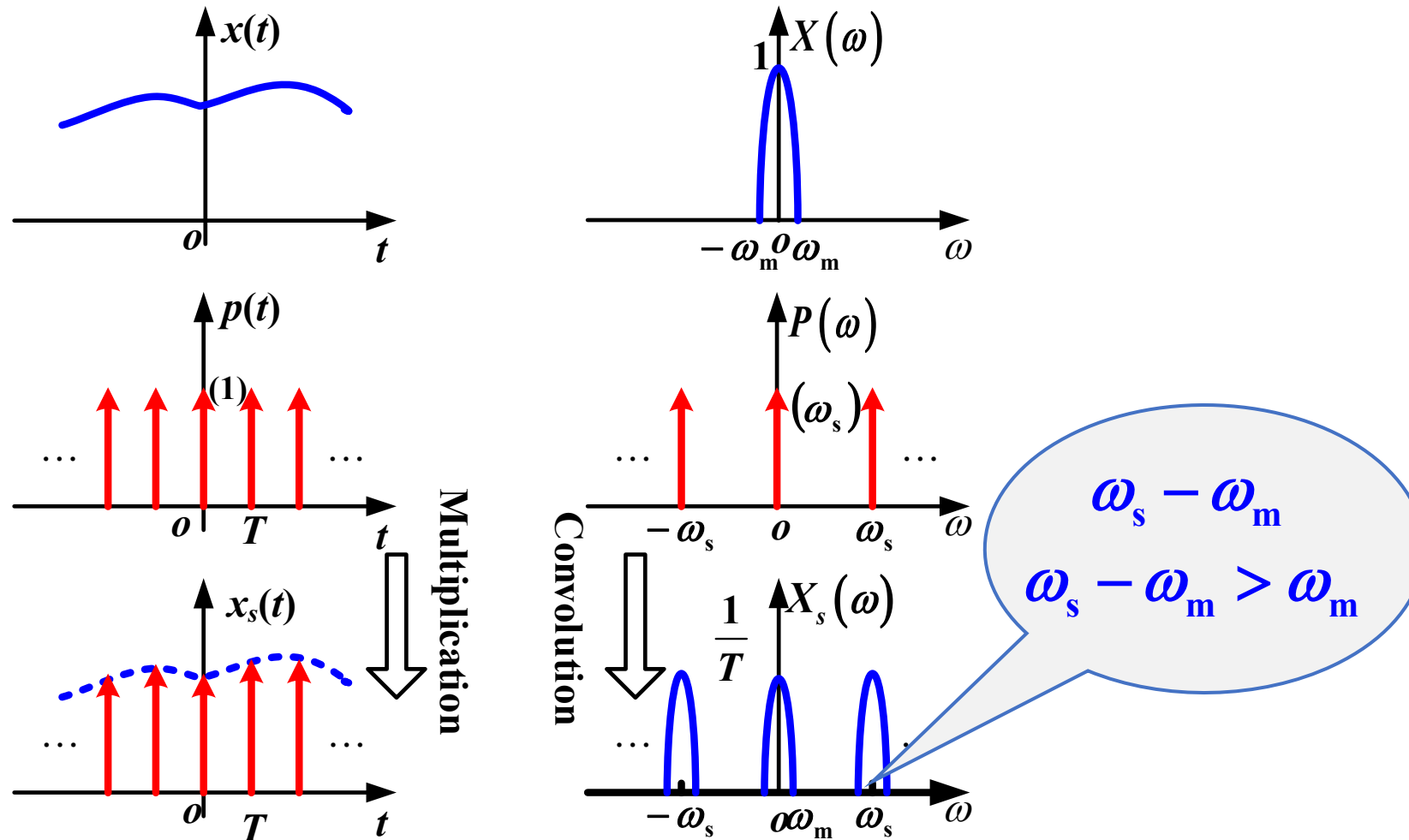
$$P(\omega) = \omega_s \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s)$$

$$X_s(\omega) = \frac{1}{2\pi} X(\omega) * \left[\omega_s \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s) \right]$$

$$= \frac{1}{T} \sum_{n=-\infty}^{\infty} X(\omega - n\omega_s)$$

4. Sampling Theorem and Aliasing

2. Spectrum of Sampled Signals



3. Sampling Theorem

- (1) For a band-limited signal $f(t)$ with a spectrum confined to $-\omega_m \sim +\omega_m$ (bandwidth: ω_m)
- (2) The signal $f(t)$ can be uniquely represented by uniformly spaced samples, provided the sampling interval T satisfies

$$T \leq \frac{1}{2f_m} \quad (\omega_m = 2\pi f_m)$$

- (3) Perfect reconstruction of $f(t)$ from $f_s(t)$ is achievable using an ideal low-pass filter with a gain of T , Cutoff frequency:

$$\omega_m < \omega_c < \omega_s - \omega_m$$

4. Sampling Theorem and Aliasing

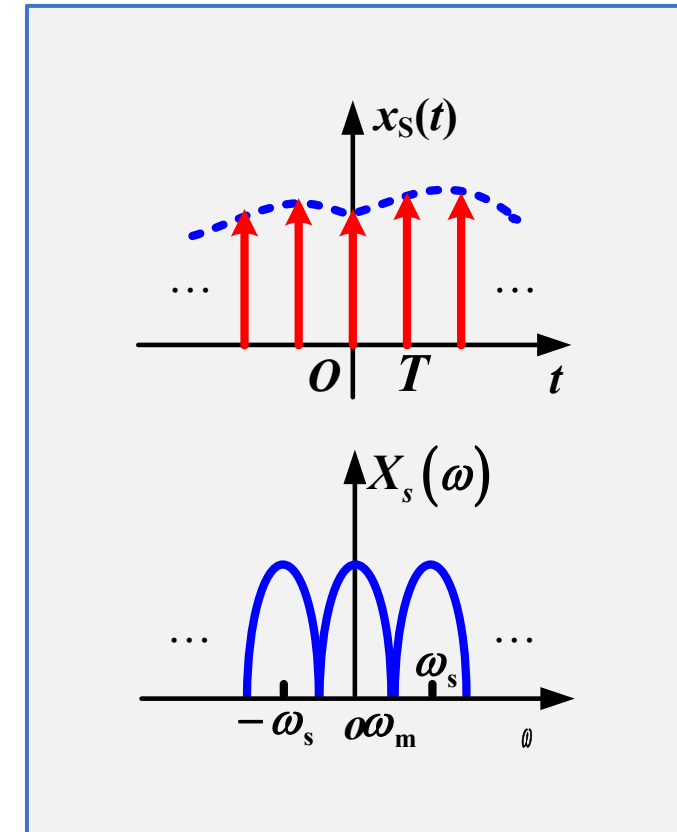
3. Sampling Theorem

$$\omega_s = 2\omega_m, f_s = 2f_m$$

Minimum allowable sampling frequency:
Known as the "Nyquist sampling frequency"

$$T = \frac{1}{2f_m}$$

Maximum allowable sampling interval:
Known as the "Nyquist sampling interval"



This nomenclature honors the pioneering contributions of **American physicist Harry Nyquist** (1889–1976) in the field of data transmission

4. Sampling Theorem and Aliasing

Example 1

For example, in audio signals, after applying an anti-aliasing filter to remove high-frequency components (spectrum range: 0–3.4 kHz), how should the sampling rate be selected?

$$f_m = 3.4 \text{ kHz}, f_s \geq 2f_m$$

$$f_{s\min} = 6800 \text{ Hz}, T_{s\max} = \frac{1}{2f_m}$$

$$\text{If let } f_s = 8000 \text{ Hz, then } T = \frac{1}{8000} = 125 \mu\text{s}$$

4. Sampling Theorem and Aliasing

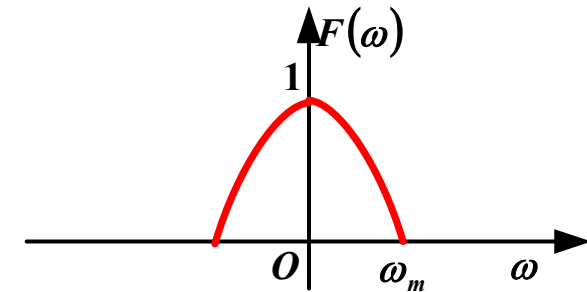
Example 2

For a band-limited signal $f(t)$ with a maximum frequency of 100 Hz, if the following signals are sampled in the time domain, determine the minimum sampling frequency $f_{\min}$

$$(1) f_1(t) = f(3t) \quad (2) f_2(t) = f(t) * f(2t)$$

$$(1) f_{1m} = 300 \text{ Hz} \quad f_{1s} \geq 600 \text{ Hz}$$

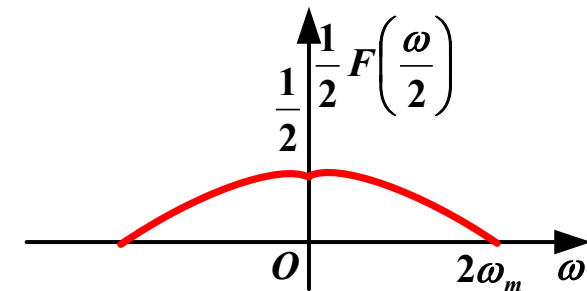
$$(2) f(t) * f(2t) \leftrightarrow F(\omega) \cdot \frac{1}{2} F\left(\frac{\omega}{2}\right)$$



$f(2t)$, bandwidth of 200 Hz

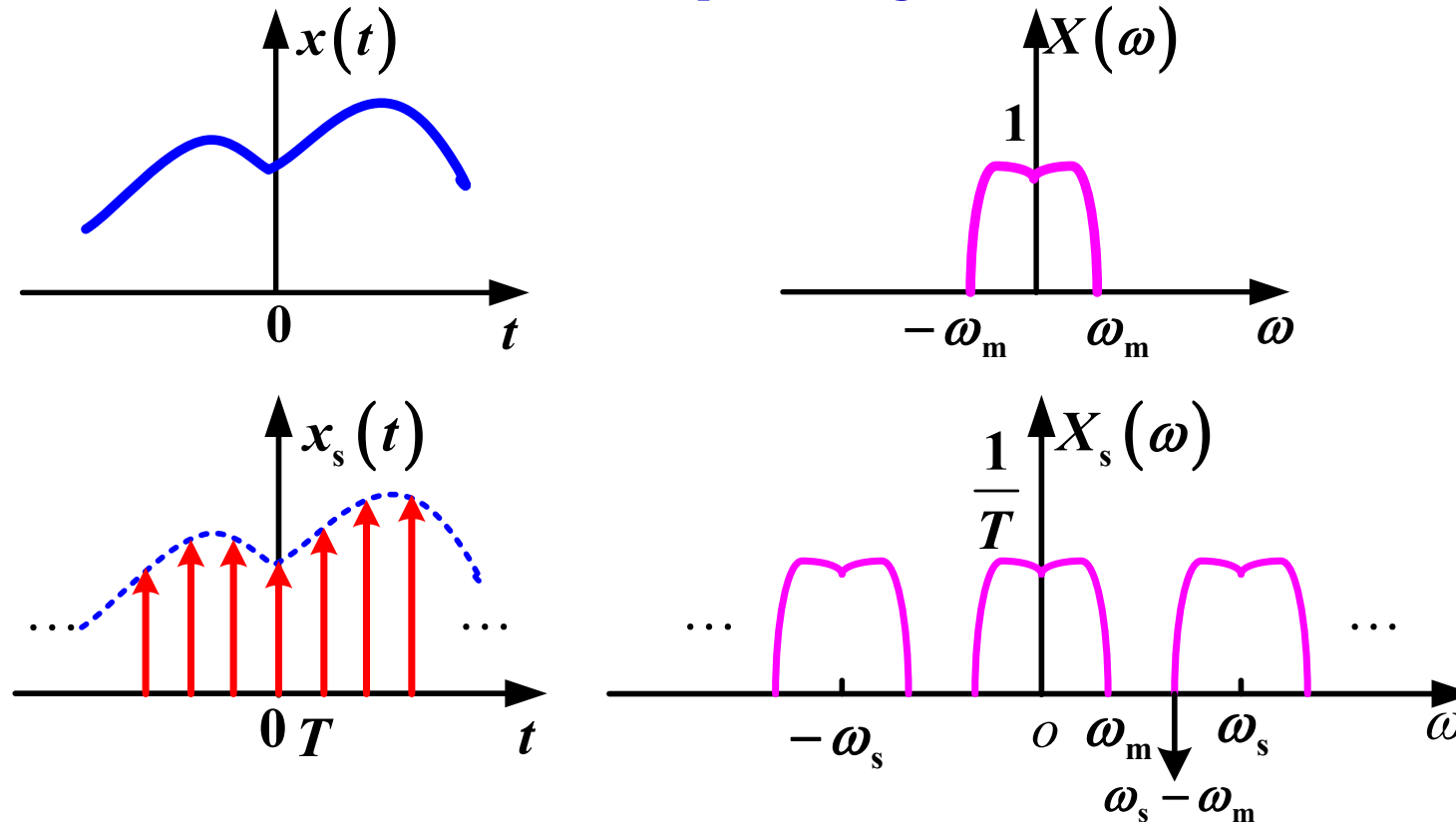
$$f_{2m} = 100 \text{ Hz}$$

$$f_{2s} \geq 200 \text{ Hz}$$



4. Sampling Theorem and Aliasing

4. Signal Reconstruction-Frequency Domain Filtering



Sampling Theorem:

- (1) Bandwidth requirement, (2) Rate requirement (2x relationship)
- (3) Recovery: Low-pass filter

4. Sampling Theorem and Aliasing

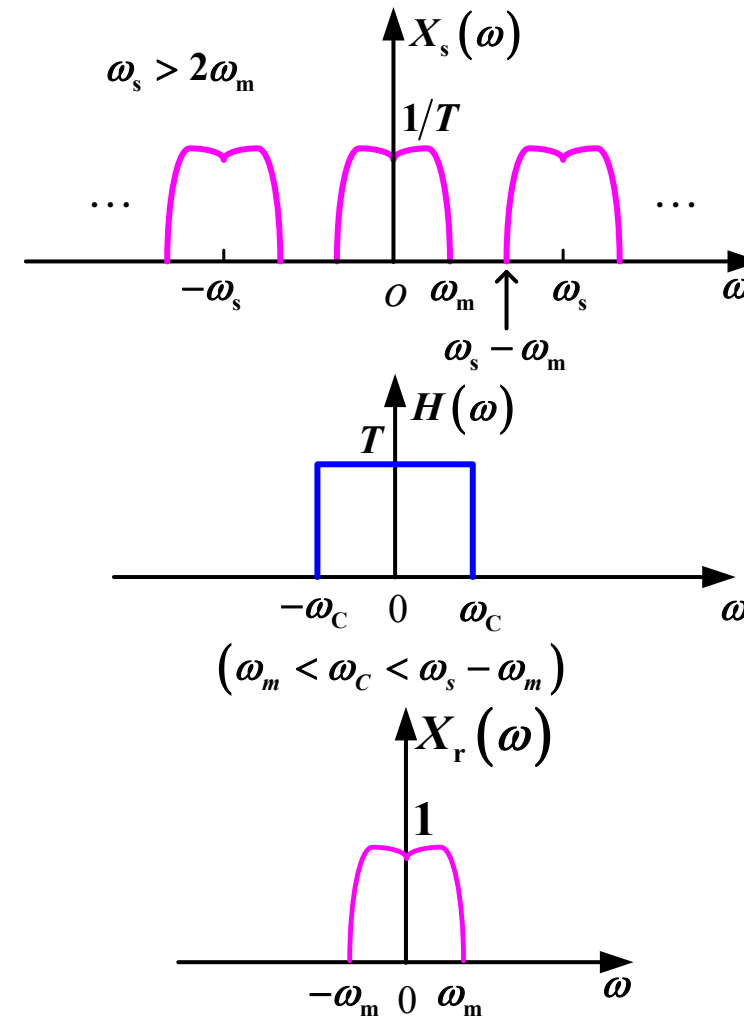
4. Signal Reconstruction-Frequency Domain Filtering

Ideal Low-Pass Filter:

$$H(\omega) = \begin{cases} T & |\omega| < \omega_c \\ 0 & |\omega| > \omega_c \end{cases}$$

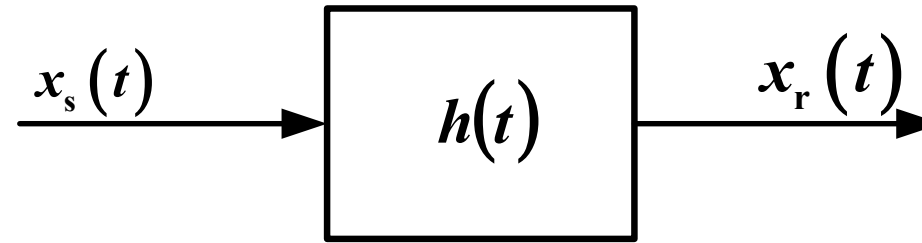
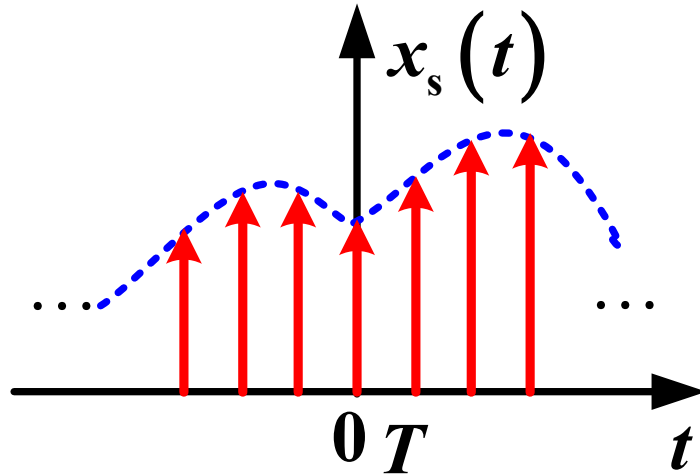
$$X_r(\omega) = X_s(\omega) \cdot H(\omega)$$

**Removing high-frequency components
reconstructs the original signal**

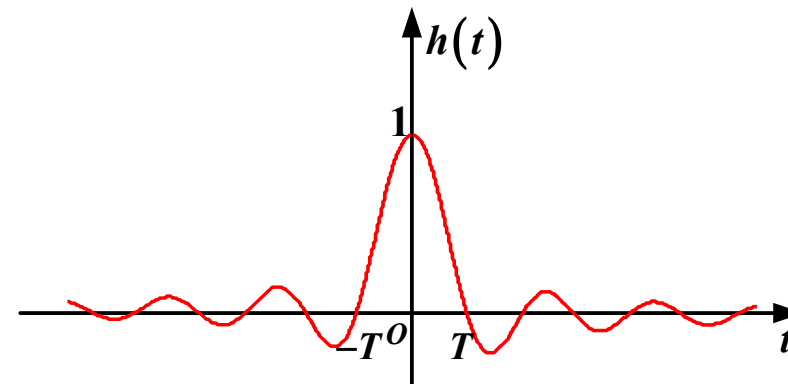


4. Sampling Theorem and Aliasing

4. Signal Reconstruction-Interpolation



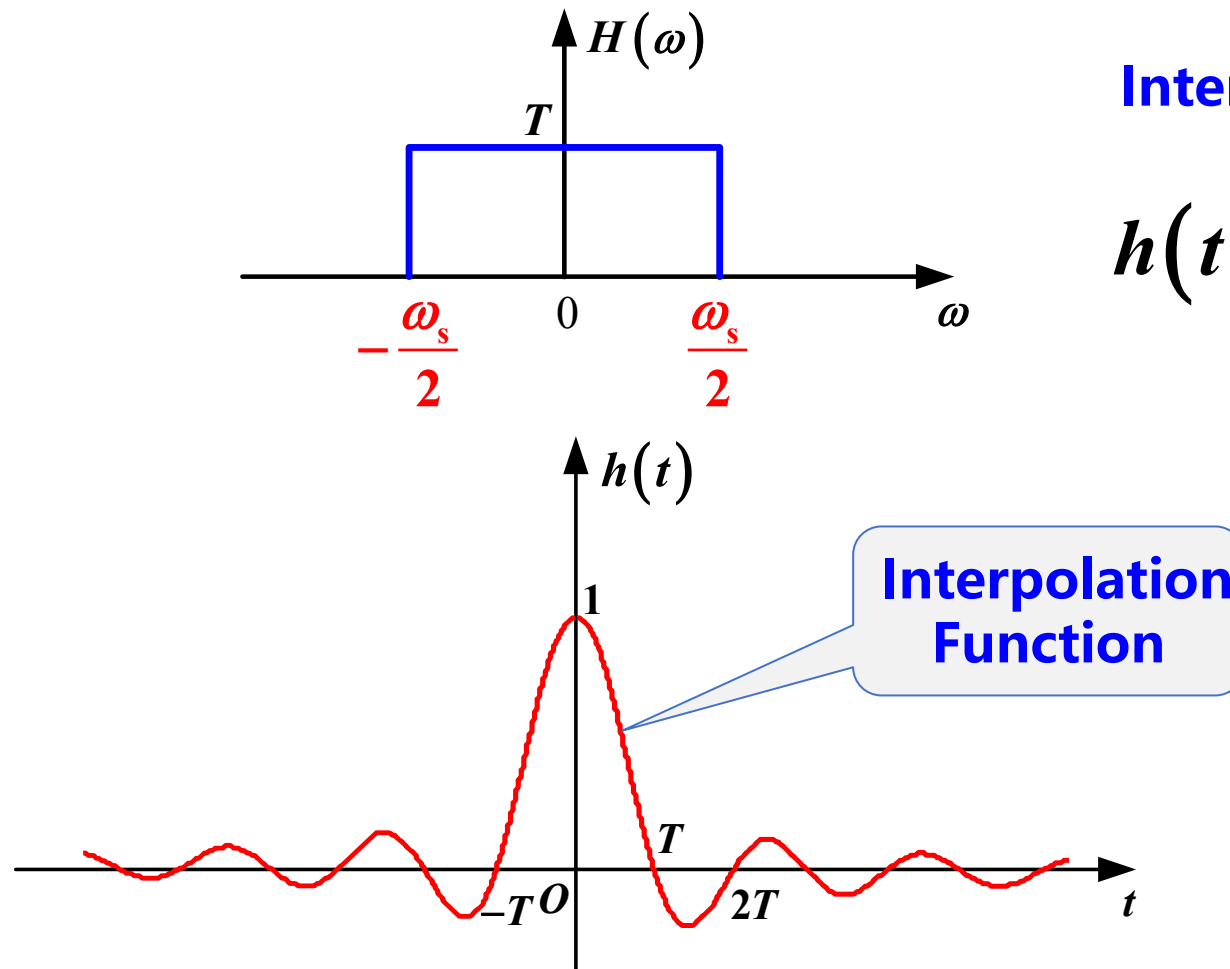
$$x_r(t) = x_s(t) * h(t)$$



Sinc function

4. Sampling Theorem and Aliasing

(1) Sa Function as Ideal Interpolation Kernel



Interpolation Function

$$h(t) = \frac{1}{2\pi} T \omega_s \text{Sa}\left(\frac{\omega_s}{2} t\right)$$
$$= \text{sinc}\left(\frac{1}{T} t\right)$$

4. Sampling Theorem and Aliasing

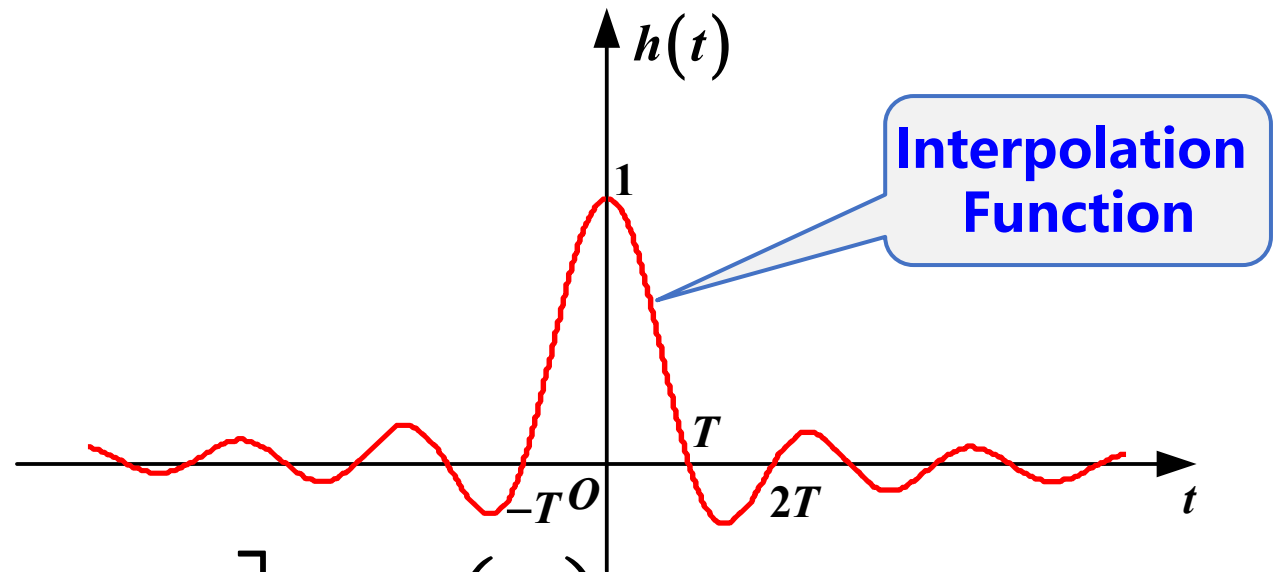
(1) Sa Function as Ideal Interpolation Kernel

$$h(t) = \text{sinc}\left(\frac{1}{T}t\right)$$

$$x_r(t) = x_s(t) * h(t)$$

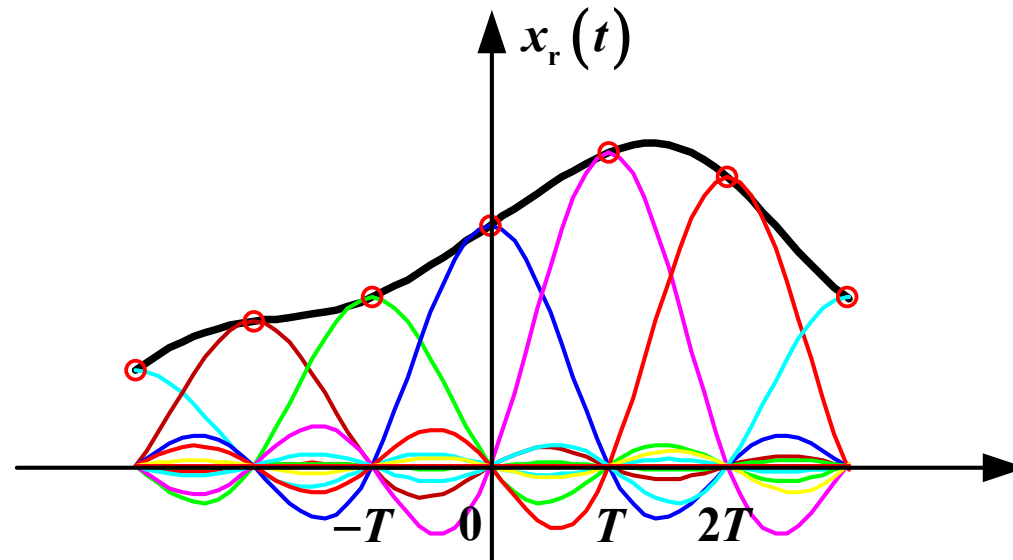
$$= \left[\sum_{n=-\infty}^{\infty} x(nT) \delta(t - nT) \right] * \text{sinc}\left(\frac{t}{T}\right)$$

$$= \sum_{n=-\infty}^{\infty} x(nT) \left[\text{sinc}\left(\frac{t - nT}{T}\right) \right]$$



4. Sampling Theorem and Aliasing

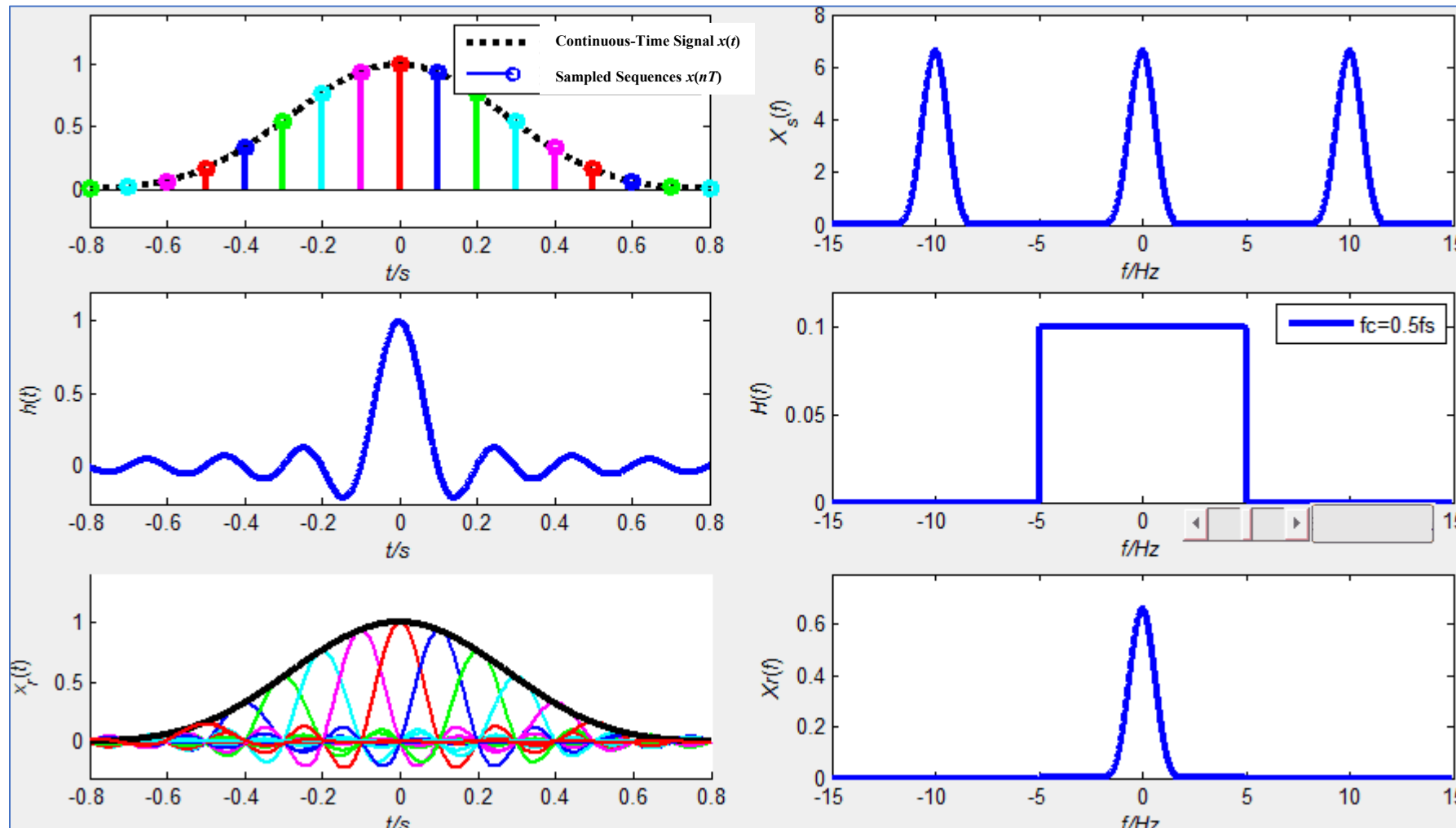
(1) Sa Function as Ideal Interpolation Kernel



- The **zeros** of the sinc interpolation function are spaced exactly at the **sampling interval T**
- **At any sampling point**, the sum of their values **equals the sample value $x(nT)$** at that point
- **At non-sampling points**, the reconstructed signal equals the superposition of multiple sinc functions, enabling **perfect recovery** of the original continuous-time signal

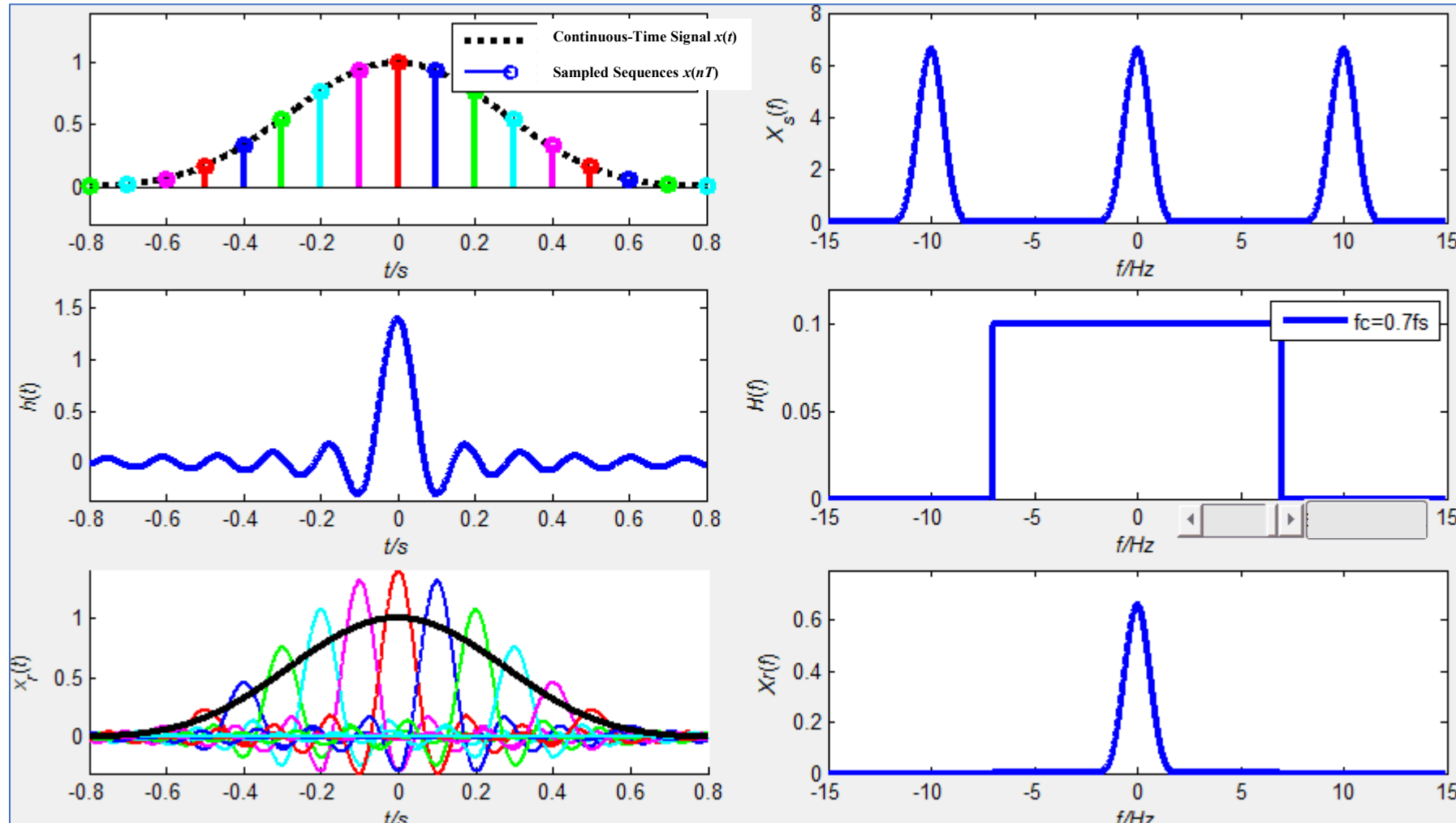
4. Sampling Theorem and Aliasing

(1) Sa Function as Ideal Interpolation Kernel $\omega_c = \frac{\omega_s}{2}$



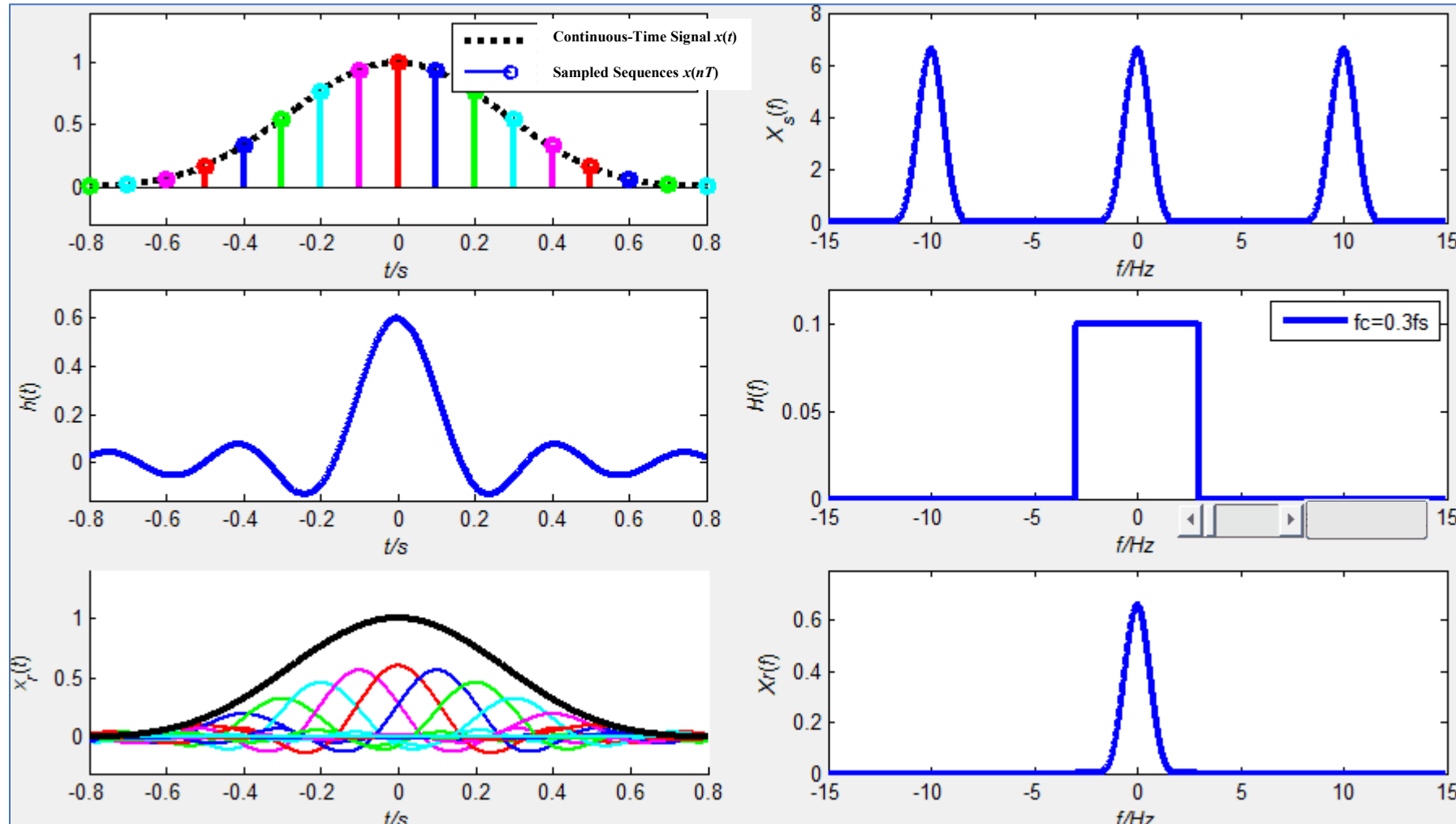
4. Sampling Theorem and Aliasing

(1) Sa Function as Ideal Interpolation Kernel $\frac{\omega_s}{2} < \omega_c < \omega_s - \omega_m$



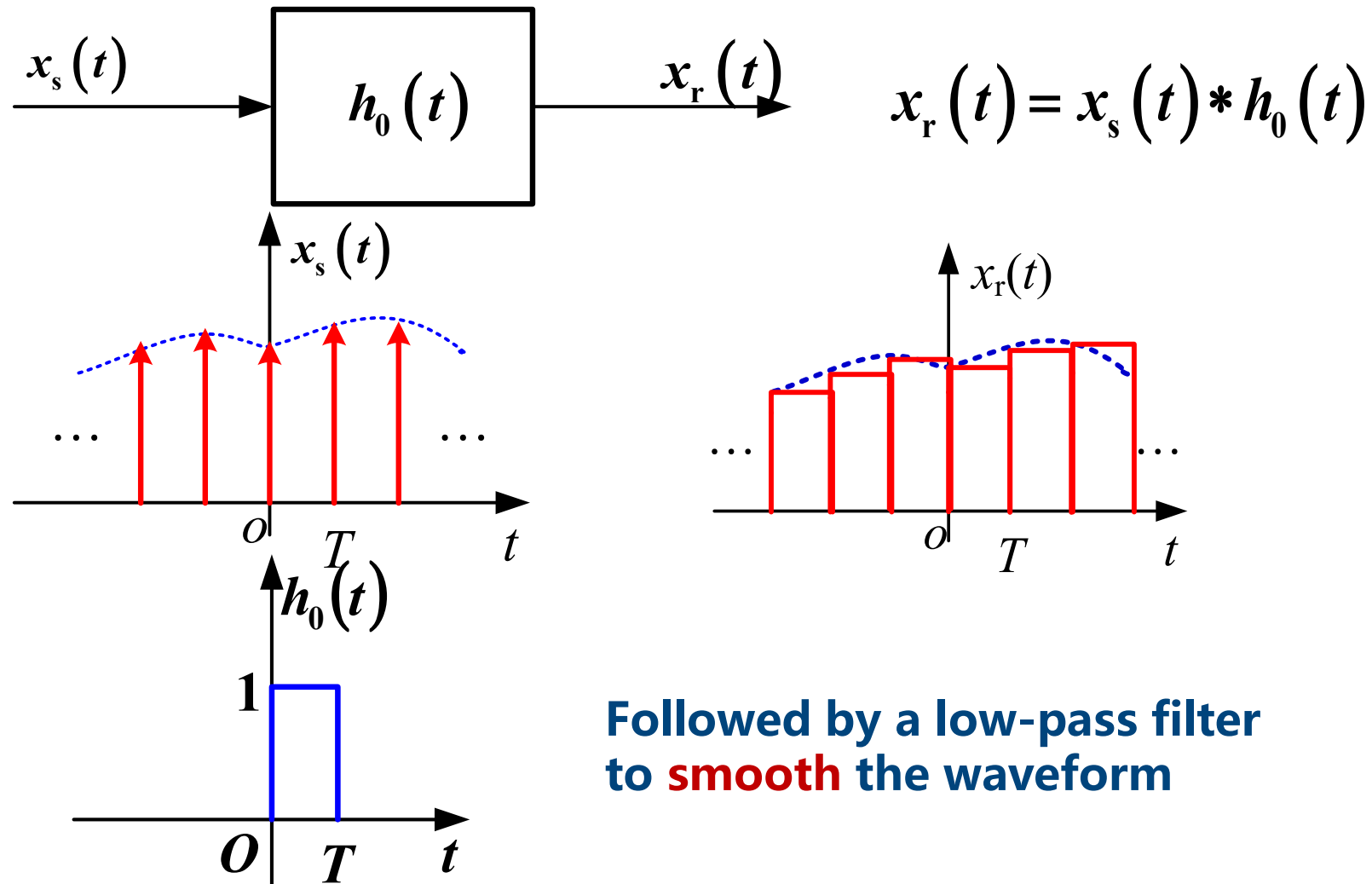
4. Sampling Theorem and Aliasing

(1) Sa Function as Ideal Interpolation Kernel $\omega_m < \omega_c < \frac{\omega_s}{2}$



4. Sampling Theorem and Aliasing

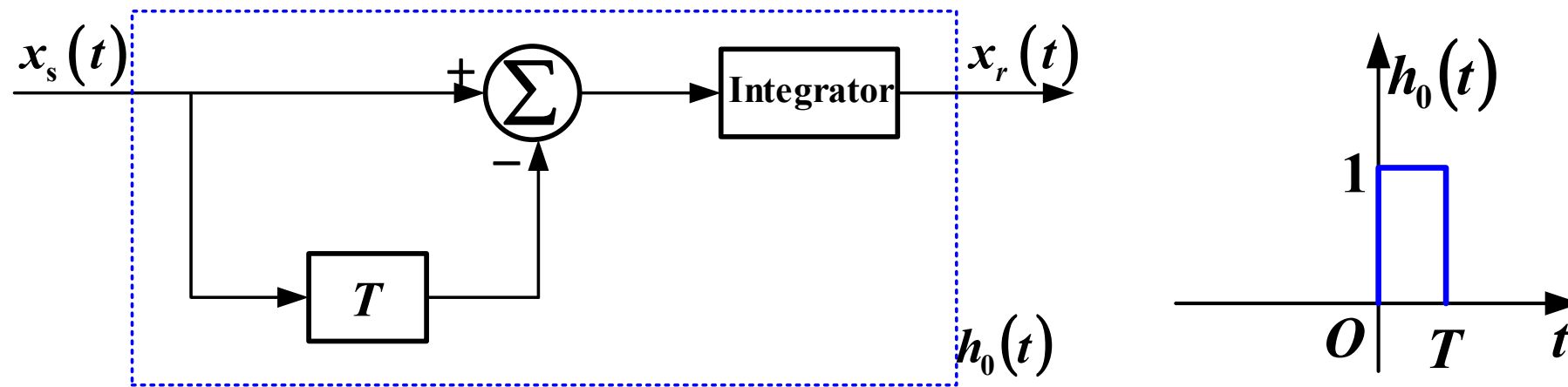
(2) Zero-order Hold (rectangular): engineering applications



4. Sampling Theorem and Aliasing

(2) Zero-order Hold (rectangular): engineering applications

Zero-Order Hold System Architecture

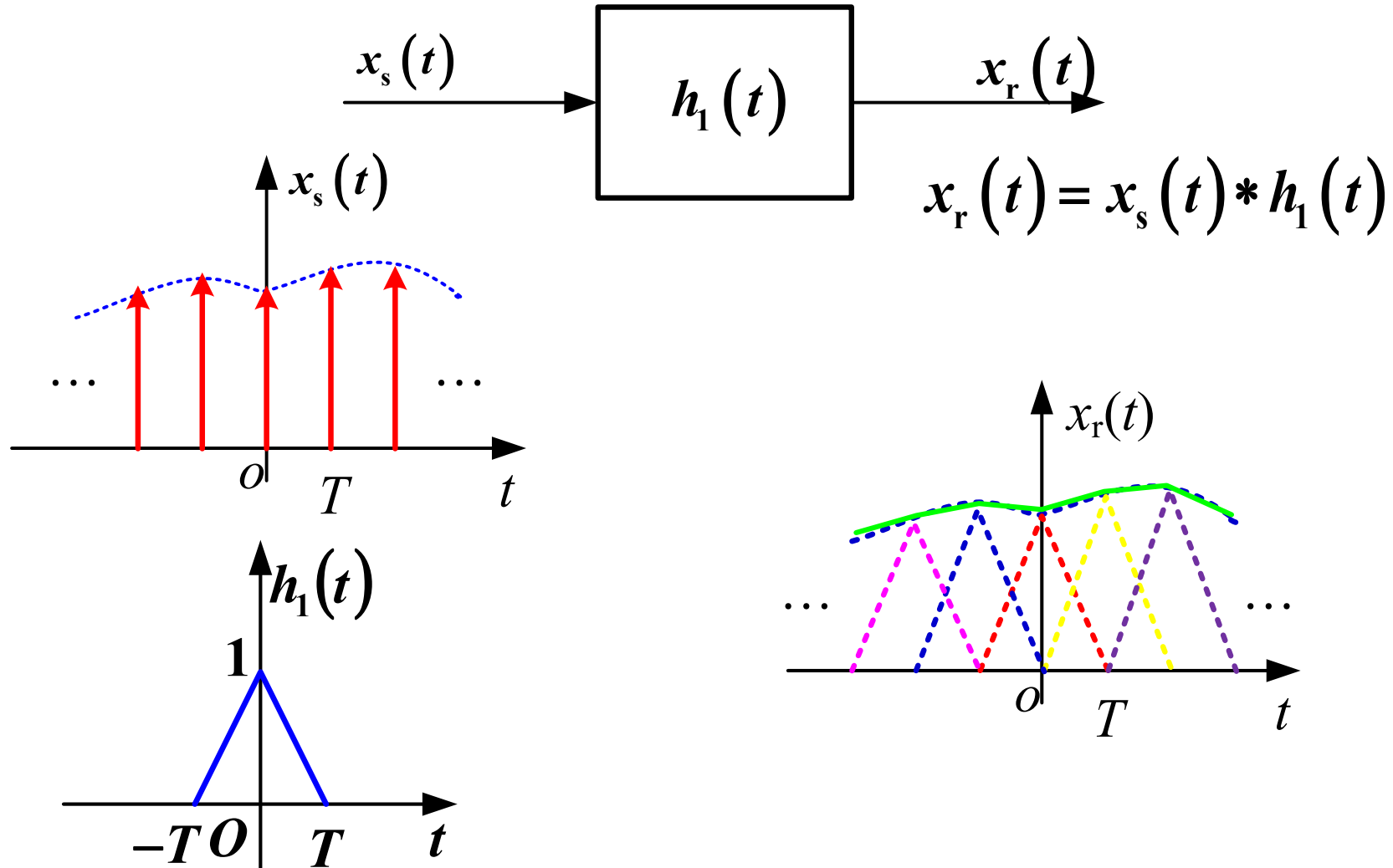


Unit Impulse Response

$$h_0(t) = \int_{-\infty}^t [\delta(t) - \delta(t - T_s)] dt = u(t) - u(t - T)$$

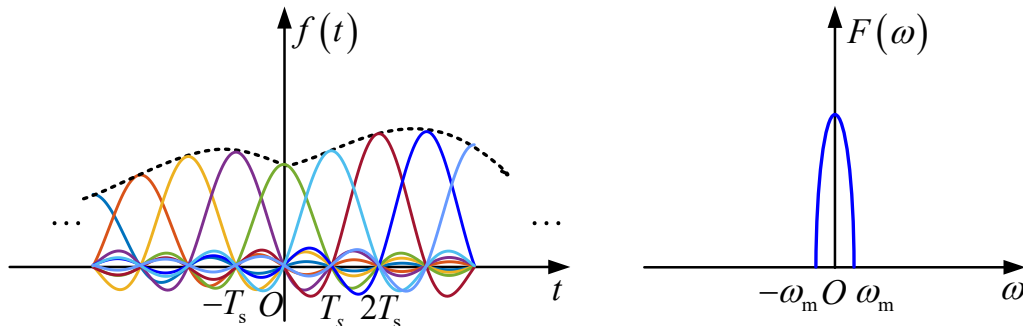
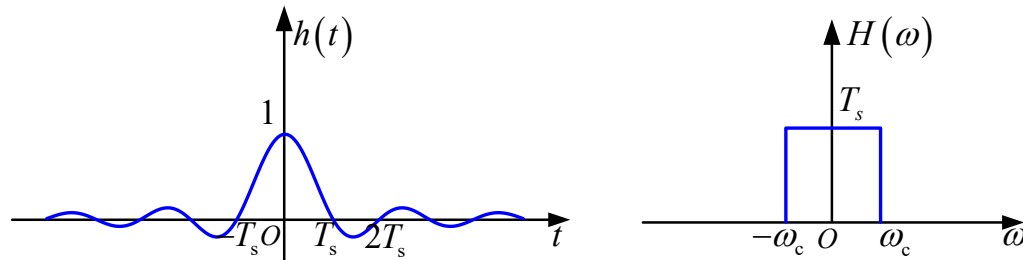
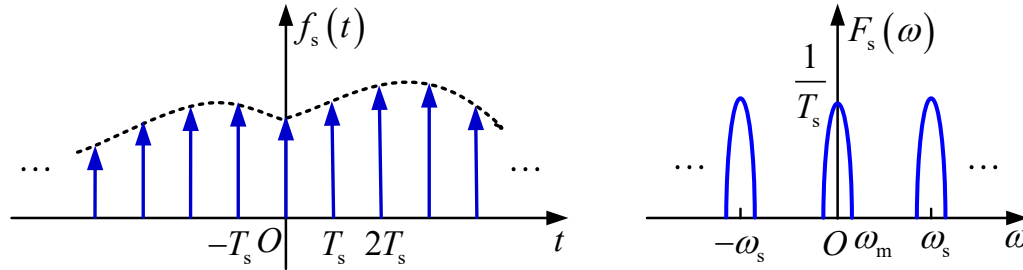
4. Sampling Theorem and Aliasing

(3) First-Order Hold (triangular)



4. Sampling Theorem and Aliasing

Summary



$$f_d(n) = f(t)|_{t=nT}$$

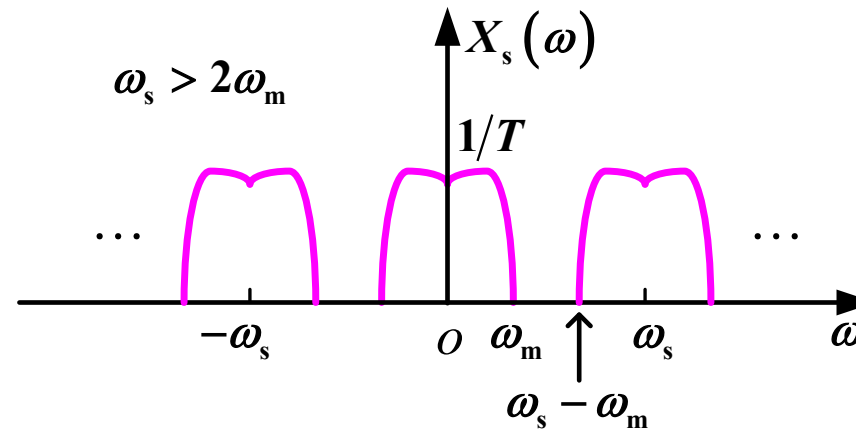
T : Sampling interval

$$f_s = \frac{1}{T} : \text{Sampling frequency}$$

Sampling Theorem: $f_s \geq 2f_m$

4. Sampling Theorem and Aliasing

5. Aliasing



$$X_s(\omega) = \frac{1}{T} \sum_{n=-\infty}^{\infty} X(\omega - n\omega_s)$$

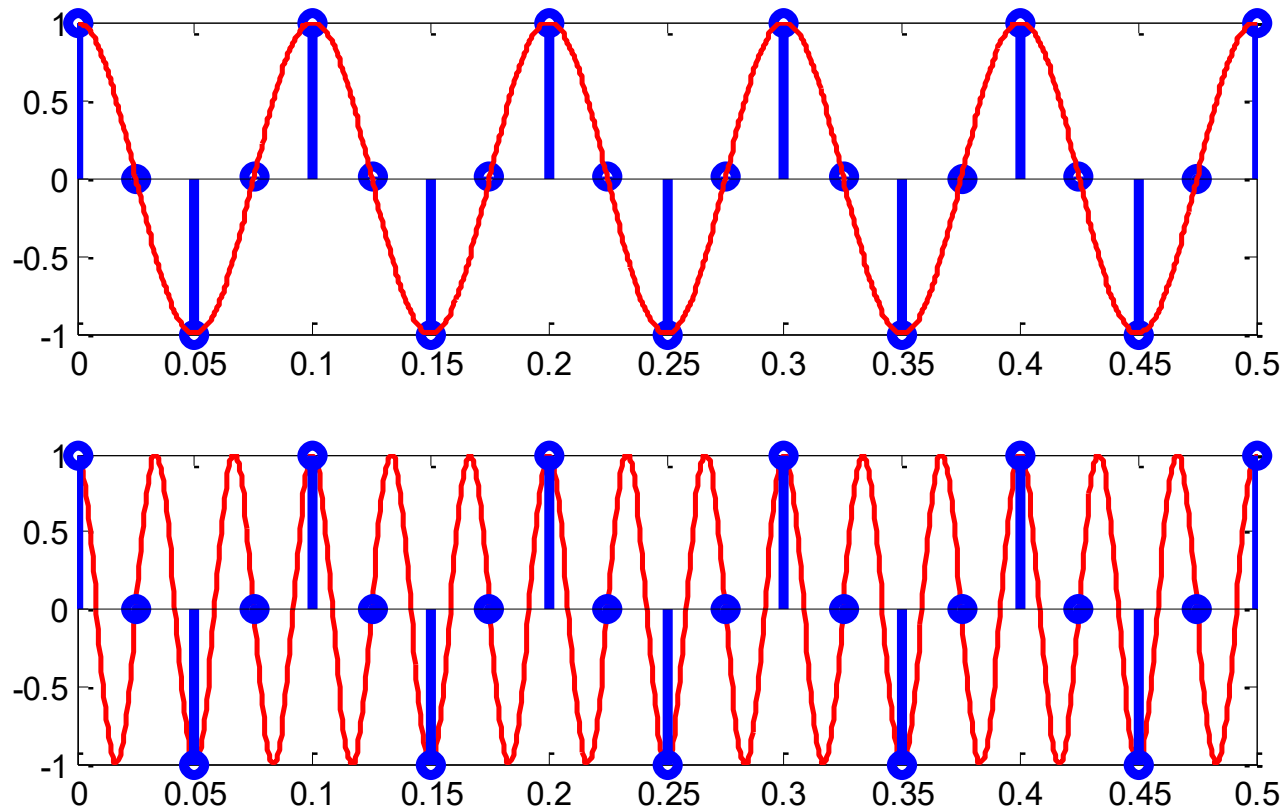
For a frequency component in real signals ω_1

Its bilateral spectrum representation consists of $\begin{cases} \omega_1 \\ -\omega_1 \end{cases}$ After Sampling: $\begin{cases} \omega_1 + n\omega_s \\ -\omega_1 + n\omega_s \end{cases}$

4. Sampling Theorem and Aliasing

Example 1

Sampling frequency is 40 Hz, sampling signals with frequencies of 10 Hz and 30 Hz. The waveforms of the sampled signals are as follows:



4. Sampling Theorem and Aliasing

Example 1

$$f_s = 40 \text{ Hz} \quad T_s = \frac{1}{f_s} = \frac{1}{40} \text{ s}$$

Continuous signal

$$x_{a1}(t) = \cos(2\pi \times 10t)$$

$$x_1(n) = x_{a1}(t)|_{t=nT_s}$$

$$= \cos\left(2\pi \times \frac{10}{40}n\right)$$

Sampled sequence

$$= \cos\left(\frac{\pi}{2}n\right)$$

$$x_{a2}(t) = \cos(2\pi \times 30t)$$

$$x_2(n) = x_{a2}(t)|_{t=nT_s}$$

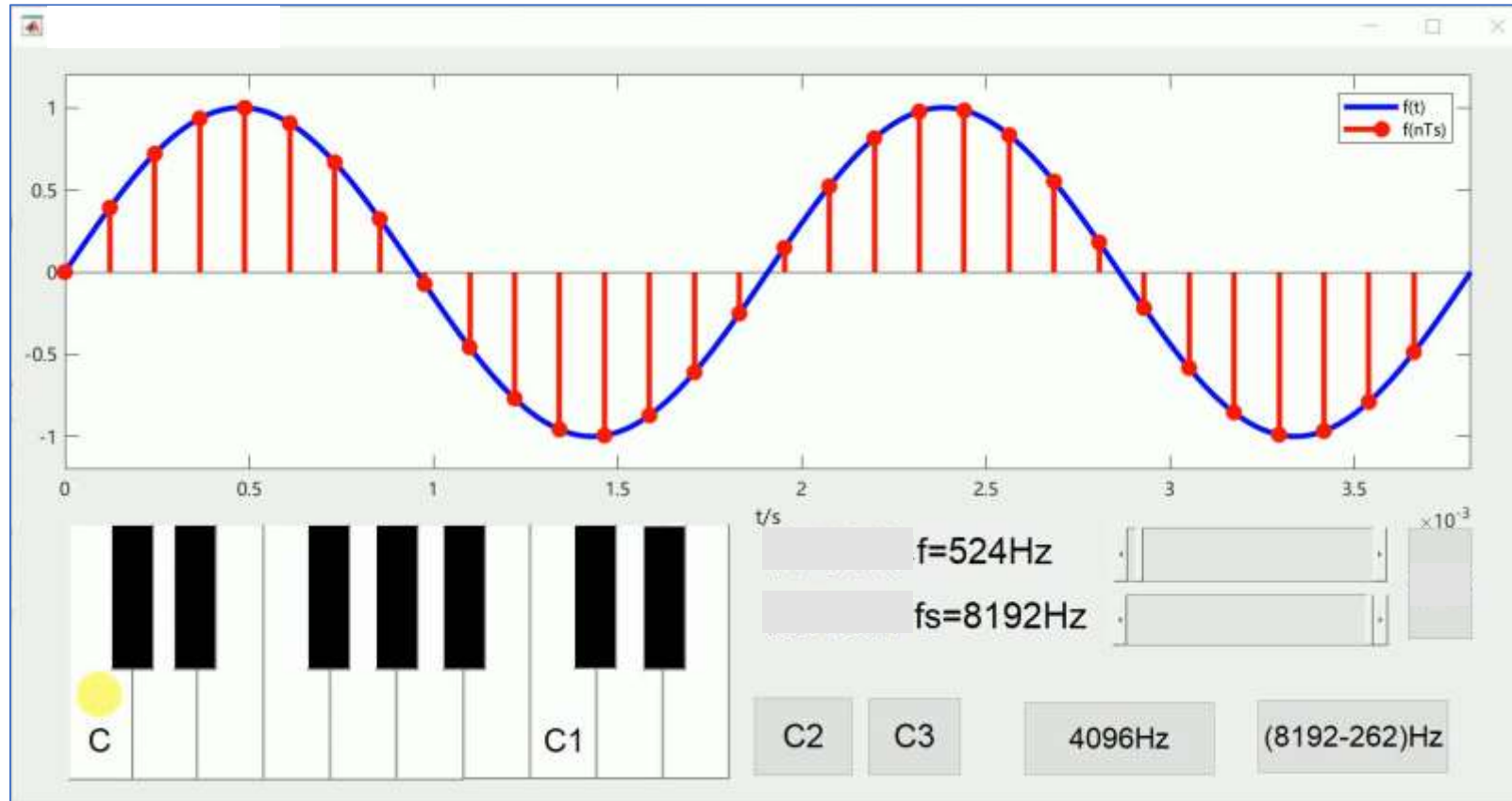
$$= \cos\left[2\pi \times \frac{30}{40}n\right] = \cos\left(2\pi n - \frac{\pi}{2}n\right)$$

$$= \cos\left(\frac{\pi}{2}n\right)$$

4. Sampling Theorem and Aliasing



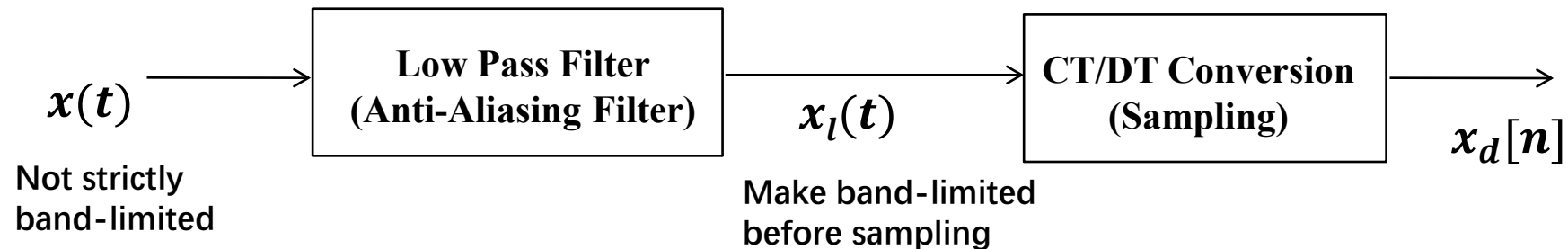
Example 1



5. Aliasing (Anti-Aliasing)

In many CT to DT conversion systems, we want to keep the sampling rate low, but the input CT signal may not be strictly band-limited to $\frac{1}{2}$ of the sampling rate

- A *anti-aliasing filter* (AAF), which is simply a low-pass filter, is often used to remove the high frequency components in the CT signal before sampling

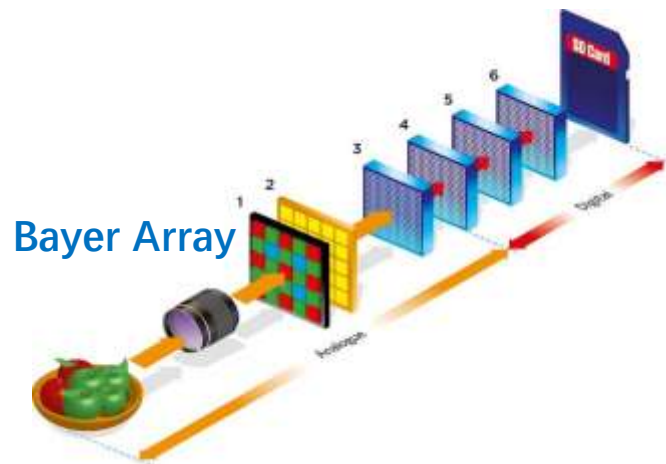


- With anti-aliasing filter, the error is *only* in the high frequency components lost
- Without anti-aliasing filter, *the high frequency components are not only lost, they turn into low frequency signals that should not be there!*

4. Sampling Theorem and Aliasing

5. Aliasing (Anti-Aliasing)

- The digital telephone network samples voice signals at 8 kHz. Therefore, filter is applied to first remove voice signal components above 4 kHz before sampling
- When we create the thumbnail of a picture, we must first low pass filter it to avoid the Moire effect
- Behind the lens of your digital camera is a filter that blurs the optical signal before spatial sampling by the CMOS sensor

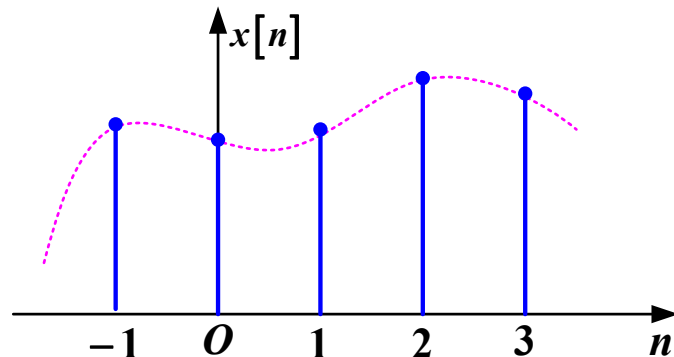
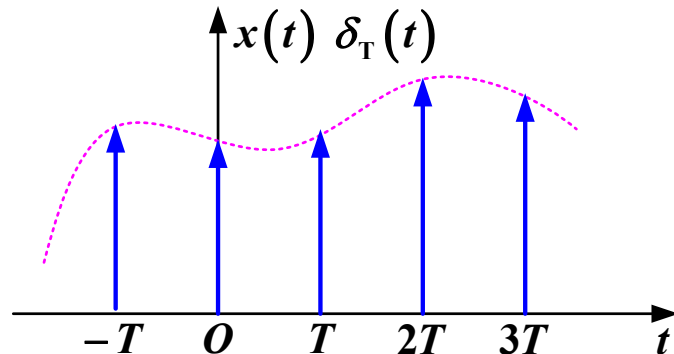


Main Content

- The Fourier Transform of a discrete-time signal (DTFT) is the Fourier transform of the sampled signal

5. Fourier Transform of a Sequence

1. Definition of DTFT



A sampled signal can be expressed as

$$\begin{aligned}x_s(t) &= x(t) \delta_T(t) \\ &= \sum_{n=-\infty}^{+\infty} x(nT) \delta(t - nT)\end{aligned}$$

Its Fourier transform

$$\begin{aligned}\mathcal{F} \left[\sum_{n=-\infty}^{\infty} x(nT) \delta(t - nT) \right] \\ = \sum_{n=-\infty}^{\infty} x(nT) e^{-jn\omega T}\end{aligned}$$

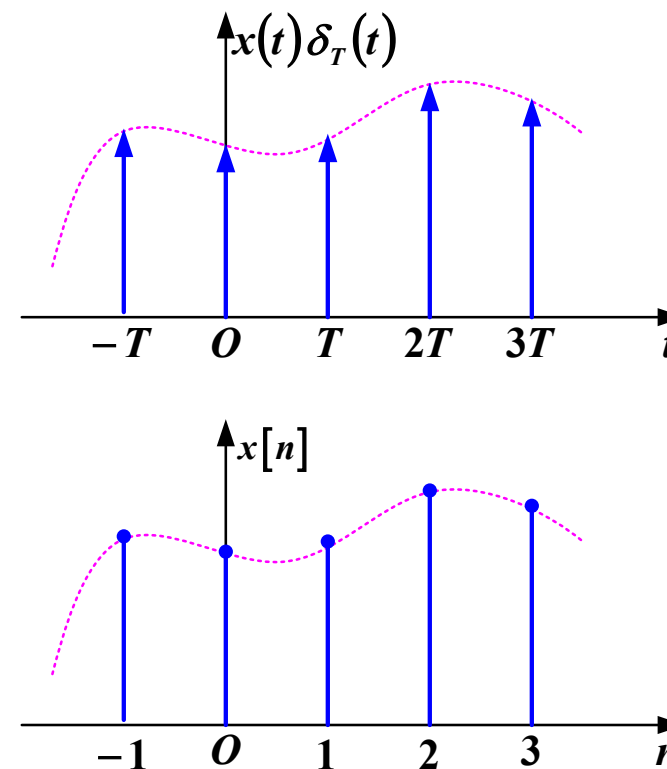
ω Analog angular frequency

5. Fourier Transform of a Sequence

1. Definition of DTFT

Let $x(nT)$ be denoted as $x[n]$, and $\omega T = \Omega$ Ω digital angular frequency

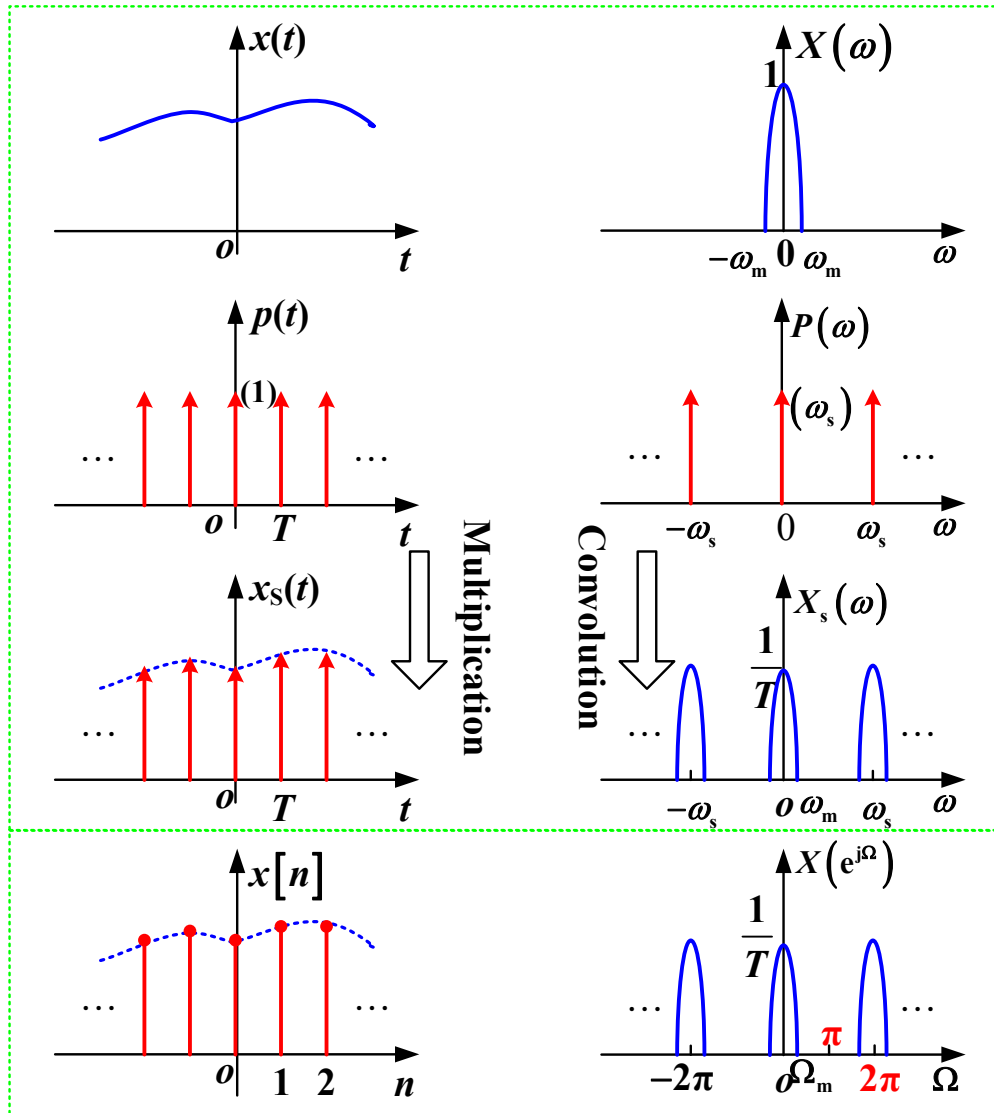
$$\begin{aligned}\mathcal{F}[x(t)\delta_T(t)] &= \sum_{n=-\infty}^{\infty} x(nT) e^{-jn\omega T} \\ &= \sum_{n=-\infty}^{\infty} x[n] e^{-jn\Omega} \\ &= \text{DTFT}[x[n]] \\ &= X(e^{j\Omega})\end{aligned}$$



This is called the Fourier Transform of a discrete-time signal
Discrete-time Fourier transform (DTFT)

5. Fourier Transform of a Sequence

1. Definition of DTFT



- DTFT is the Fourier Transform of a Sampled Signal
- DTFT has a period of 2π

$$\Omega = \omega T$$

$$\text{when } \omega = \omega_s$$

$$\Omega = \omega_s T = 2\pi$$

5. Fourier Transform of a Sequence

1. Definition of DTFT

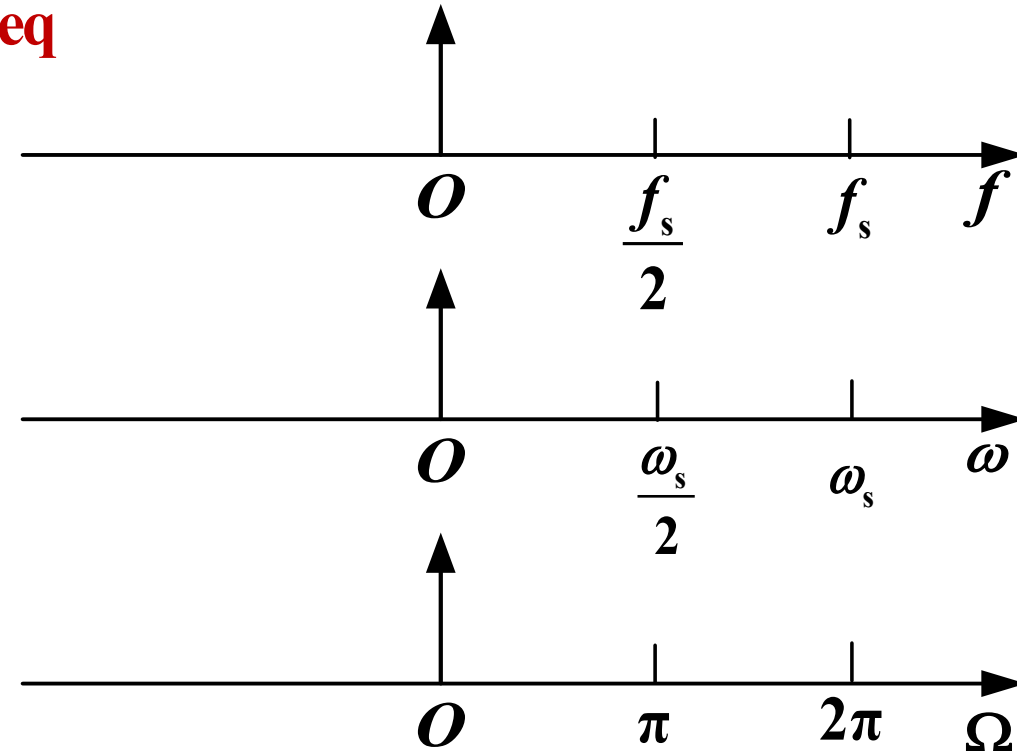
Frequency, Analog Ang Freq, and Dig Ang Freq

Frequency f

Analog Angular Frequency $\omega = 2\pi f$

Digital Angular Frequency

$$\Omega = \omega T = \frac{\omega}{f_s} = 2\pi \frac{f}{f_s}$$



Given a sampling frequency f_s , the **highest** frequency of the signal being processed is $\frac{f_s}{2}$:

$$\Omega_m \leq \pi$$

This means π is the maximum value of the digital angular frequency

1. Definition of DTFT

□ Using the frequency-domain convolution theorem

$$X_s(\omega) = \frac{1}{T_s} \sum_{n=-\infty}^{+\infty} X(\omega - n\omega_s)$$

This indicates that the spectrum of the sampled signal is a periodic extension of the original signal's spectrum

□ Using the time-shift property (DTFT)

$$X(e^{j\Omega}) = \sum_{n=-\infty}^{+\infty} x[n]e^{-jn\Omega}$$

This provides a method that can be used in engineering to compute the frequency spectrum